

Generation-Time Routing Reveals Expert Specialization That Prefill Measurements Miss

Jeffrey Shorthill*

Abstract

In two Qwen3.5 mixture-of-experts (MoE) checkpoints (35B and 122B parameters) run on 60 prompts spanning twenty academic domains, the expert that receives the most routing weight in a domain depends on which regime of the forward pass is measured. Over prefill, when the prompt is processed in parallel, one generalist expert wins 18 of 20 domains in the smaller model and 13 of 20 in the larger, and routing looks domain-blind. Over generation, when the answer is produced token by token, the winners disperse to 20 and 18 distinct experts and the generalist wins one or two. The dispersion is a property of the regime: with prefill and generation matched in token count, the two regimes select nearly disjoint top-expert sets (Jaccard 0.00 to 0.23) at unchanged per-token routing entropy (0.958 versus 0.953), so generation spreads each token over a similar number of experts and routes it to different ones, and the pattern reproduces across model sizes with unrelated expert indices. The widely cited negative result that MoE experts do not specialize by topic was measured over supplied text, which is prefill routing. Expert-specialization measurements over supplied text, or pooled across regimes, are therefore understated by construction; prefill-based negative results are silent about the regime in which specialization is expressed; and analyses that ask what experts specialize in should separate the two regimes and read specialization from generation.

1 Introduction

Mixture-of-experts (MoE) layers replace a dense feed-forward block with many parallel expert blocks and a router that selects a few of them for each token. Because the router makes a discrete, inspectable choice at every token, it invites a natural question: do the experts divide knowledge by subject? The most-cited answer is

negative. Jiang et al. (2024), analyzing Mixtral’s routing on validation text, “do not observe obvious patterns in the assignment of experts based on the topic”; the structure they find is syntactic and positional. That result is load-bearing: it is widely read as evidence that MoE experts do not specialize by domain.

That analysis, and most like it, scored routing over supplied text, and supplied text is processed in only one of the two regimes in which a transformer routes tokens. During *prefill*, the whole prompt is processed in parallel to populate the key/value cache. During *generation*, output tokens are produced one at a time, each attending to everything before it. Prefill routing is the natural place to look: it is deterministic given the prompt and the weights, cheap to capture in a single forward pass, and it is where the prompt’s content sits, so topical routing is intuitively expected to be strongest there. Routing happens in both regimes, however, and nothing requires them to agree.

They disagree in a specific, consistent way that decides whether domain specialization is visible. On a domain-balanced prompt suite run through two MoE models, the expert that receives the most routing weight in each of twenty domains is a single generalist for 18 of 20 domains in the smaller model and 13 of 20 in the larger when routing is read over prefill positions, and a distinct expert for nearly every domain when the same prompts are read over generated positions. The dispersion survives a control that matches prefill and generation in token count, and it reproduces in the larger model with unrelated expert indices. Prefill routing reproduces the familiar domain-blind picture; generation routing resolves into per-domain experts.

The scope is *where* in the forward process domain structure becomes measurable; what each expert encodes, and a re-measurement of Mixtral, are left to separate work. The result carries a methodological correction. Reading expert specialization off prefill or supplied-text routing, or off routing pooled across regimes without separation, systematically understates it. A negative result obtained over supplied text reports the prefill regime accurately and leaves the generation regime unmeasured; when the question is what an MoE’s experts specialize in, the two regimes should be measured separately and specialization read from

*Independent researcher. Correspondence: jws299792@icloud.com. Version 1.1 (August 2026; version 1.0 June 2026), released under CC BY 4.0. Concept DOI 10.5281/zenodo.20779604. Data, code, and a claim-by-claim source index accompany the paper; see Data and Code Availability.

generation.

2 Background

Mixture-of-experts routing. The MoE construction routes inputs to specialized subnetworks through a learned gate (Jacobs et al., 1991); Shazeer et al. (2017) made it sparse and scalable with a top- k gate over thousands of experts, and the design was carried into large transformers by Switch Transformers (Fedus et al., 2022) and ST-MoE (Zoph et al., 2022). The models studied here are open MoE checkpoints in the Qwen3 lineage (Yang et al., 2025).

Do experts specialize by topic? Jiang et al. (2024) report that Mixtral’s expert distribution is similar across arXiv, biomedical, and philosophy documents, with structure in syntax and token position rather than topic. The statement is carefully scoped to what it measured, routing over supplied text, and it has since hardened in looser uptake into the belief that MoE experts do not specialize by domain. Dai et al. (2024) reach a compatible intuition from the engineering side, treating strong specialization as something to build in through fine-grained expert segmentation and shared-expert isolation, which presumes that ordinary top- k routing specializes weakly. Our measurement extends the first result and qualifies the second: read over prefill we also see little domain structure, and the same probe read over generation resolves into per-domain experts. The difference from Jiang et al. (2024) is the axis of measurement, the forward-pass regime.

Input-dependent routing. Routing depends strongly on the input. Avinash (2026) recover task identity from routing alone with high accuracy, and ichiro Hayashi et al. (2026), working with the same Qwen3.5-35B-A3B checkpoint we use, find that routing during code generation is strongly token- and layer-dependent. Both condition on what is being processed and pool within a regime; the prefill-versus-generation contrast for the same prompts is the comparison we add. The two-regime structure of transformer inference is well established on the systems side, where the phases have distinct compute profiles and are increasingly served separately (Kwon et al., 2023; Zhong et al., 2024); we use the same distinction representationally.

3 Methods

Probe. The prompt suite contains 60 short questions covering 20 academic domains (archaeology, biology,

chemistry, comparative religion, computer science, cybersecurity, economics, environmental science, history, law, linguistics, mathematics, medicine, neuroscience, philosophy, physics, political science, psychology, software engineering, statistics), three per domain. Each is a plain, domain-typical prompt, for example a request to contrast two core concepts in the field.

Models. We use two MoE checkpoints from the Qwen3.5 lineage (Yang et al., 2025), both 8-bit quantized community fine-tunes (HauhausCS variants): a 35B-parameter model with about 3B parameters active per token (Qwen3.5-35B-A3B) and a 122B-parameter model with about 10B active (Qwen3.5-122B-A10B). Both route each token to the top 8 of 256 experts per MoE layer. The two models have different layer counts and internal layouts, so their expert indices are unrelated: an expert numbered 114 in one has nothing to do with expert 114 in the other. Decoding is greedy with no reasoning trace. Generation is capped at 2,048 to 2,056 new tokens; we return to the cap in Section 6.

Routing reconstruction. For each token we take the router’s pre-softmax logits over all 256 experts, apply a softmax, keep the top 8, and renormalize so the eight weights sum to one. For an expert e on a fixed set of tokens (one regime of one prompt at one layer) we form three reductions of these weights: S_e , the fraction of tokens on which e is selected; Q_e , its mean weight on tokens where it is selected; and W_e , its mean weight over all tokens (zero where unselected). Because all three reduce the same weights, $W_e = S_e Q_e$ holds exactly on any single token set; we use this as a bookkeeping check that the selection mask, conditional mean, and marginal mean are mutually consistent ($\max_e |W_e - S_e Q_e| \leq 5.6 \times 10^{-17}$ across all capture cells). The identity holds per token set only; the tabulated per-domain S and Q in the accompanying data are separate means whose product differs from the tabulated W by the across-prompt covariance. Domain winners are ranked by W , the mean routed weight per token.

The two regimes and the domain winner. For each prompt we partition its token positions into the prefill block (the prompt tokens) and the generation block (the tokens the model produced). All routing statistics are computed within one block, never pooled across them. For each domain we form the per-domain mean of W_e over the domain’s three prompts and all MoE layers and call the highest- W expert the *domain winner*. This yields, per model and per regime, a list of 20 domain winners; the analysis concerns how the two lists differ. We call the expert that wins the most domains in a list the *generalist*.

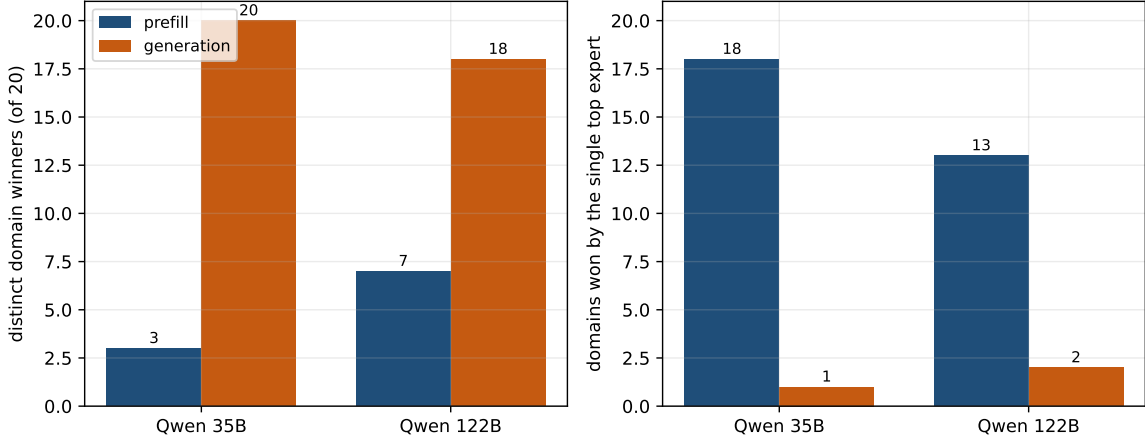


Figure 1: Left: number of distinct experts among the 20 domain winners. Right: number of domains won by the single most frequent winner. In both models the prefill winners collapse onto a generalist and the generation winners spread across the domains. $n = 60$ prompts per model (20 domains $\times$ 3).

Model	Regime	Distinct winners	Max wins	Herfindahl	Norm. entropy
Qwen 35B	prefill	3	18	0.815	0.13
Qwen 35B	generation	20	1	0.050	1.00
Qwen 122B	prefill	7	13	0.445	0.42
Qwen 122B	generation	18	2	0.060	0.95

Table 1: Concentration of the 20-domain winner list, by model and regime. Distinct winners and max wins are counts out of 20 domains; Herfindahl and normalized entropy are computed on the distribution of domains won per expert (Section 3). Each row summarizes one domain-winner table from the source archive ($n = 60$ prompts per model, 3 per domain).

Concentration of a winner list. A list of 20 winners is summarized by its *distinct count* (how many different experts appear), its *max wins* (domains taken by the single most frequent winner), the Herfindahl index $\sum_i p_i^2$, and the normalized Shannon entropy $H = -\sum_i p_i \log_2 p_i / \log_2 20$, where p_i is the fraction of domains won by expert i ; H runs from 0 (one expert wins every domain) to 1 (every domain has its own winner). These are deterministic summaries of the winner lists.

Token-balanced control. To separate the regime from token count, three prompts were each packed with twenty domain questions and padded to exactly 446 prompt tokens, and each was allowed 2,048 generated tokens, so the prefill and generation blocks are matched in scale and per-chunk comparisons use identical token budgets. For each chunk we report the Jaccard index (intersection over union) between the prefill and generation top-expert sets, ranked by W and by S , and the mean per-token routing entropy of each regime.

4 Results

4.1 Over prefill one generalist wins almost every domain

In the 35B model the prefill winner list is dominated by a single expert that takes 18 of the 20 domains; only chemistry and linguistics have their own prefill winners. The same expert is the top expert overall on the prefill block. The list has 3 distinct experts and normalized entropy 0.13 (Table 1). Read literally, prefill routing assigns one high-frequency default to nearly every subject. In the 122B model a different generalist wins 13 of 20 domains, with a second expert taking 2 more, for 7 distinct winners and entropy 0.42: less extreme, and still strongly concentrated.

4.2 Over generation the winners disperse to distinct per-domain experts

The generation winner list for the same 35B model and prompts diverges sharply. All 20 domains are won by 20 distinct experts; no expert wins more than one domain, and normalized entropy rises from 0.13

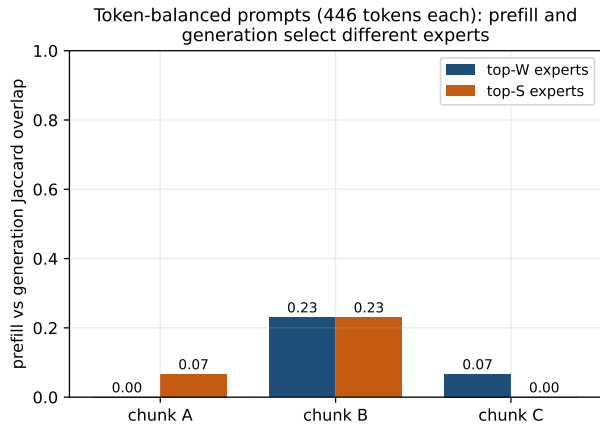


Figure 2: Token-balanced control (35B model, three packed prompts of 446 prompt tokens and 2,048 generated tokens each). Jaccard overlap between the prefill and generation top-expert sets, ranked by routed weight and by selection rate. With token counts matched, the two regimes select largely disjoint experts.

to 1.00. The generalist that won 18 domains during prefill wins exactly one during generation (political science). Domain-specific experts surface in its place: the philosophy generation winner, for instance, held only rank 5 within philosophy during prefill ($W = 0.0188$ over generation against a mid-pack prefill weight) and wins no other domain. The 122B model shows the same dispersion, to 18 distinct winners across the 20 domains (two experts win two domains each), entropy 0.95, and its global top experts also change identity between regimes. The generalist’s reach falls from 18 to 1 domains in the 35B model and from 13 to 2 in the 122B model (Figure 1).

The dispersion holds regardless of where the generated text is cut. These checkpoints render the end-of-turn marker as literal text and continue into synthetic follow-up turns until the token cap. Trimming the generation block at that marker or leaving it untrimmed produces the same 20 distinct domain winners in the 35B model. The structure sits in the answer content.

4.3 The shift persists at matched token count

The dispersion is a regime effect. Prefill and generation differ in length (a prompt is tens of tokens and an answer hundreds), so the token-balanced control asks whether longer token sets alone could disperse the top experts. With prefill and generation matched at 446 and 2,048 tokens per chunk, the Jaccard overlap between the top-expert sets of the two regimes is 0.00, 0.23, and 0.07 across the three chunks by routed weight and 0.07, 0.23, and 0.00 by selection rate (Figure 2):

the experts that dominate generation are mostly disjoint from those that dominate prefill at matched token count. Per-token routing entropy is almost unchanged between regimes (0.958 prefill, 0.953 generation), so generation spreads each token over a similar number of experts and routes it to different ones. The regime moves the winners while length is held fixed.

4.4 The pattern reproduces with unrelated expert indices

The 122B model is an independent replication in the sense that matters: its expert indices are unrelated to the 35B model’s, its generalist is a different expert, and its prefill leaders and generation leaders are different sets of indices, yet the prefill-concentrated, generation-dispersed shape carries over (Table 1). The two models also show that an expert’s role can move between regimes within a model and that indices are model-specific: in the 122B model the expert leading philosophy generation is mid-pack in philosophy prefill, and the index that wins philosophy generation in the 35B model has, in the 122B model, its only strong footprint in computer science. We read these as cautions against naming experts from any single regime or carrying an index across models.

5 Discussion

What the measurement establishes. In these models routing over prompt positions is dominated by a few high-frequency experts and carries little domain signal, while routing over generated positions resolves into distinct per-domain leaders. Domain specialization, to the extent the router expresses it, is a property of the generation regime. A deflationary reading would credit the dispersion to generation routing more diffusely in general; the two entropies measured together in the token-balanced control tell against it. Per-token routing entropy holds nearly steady across regimes (0.958 against 0.953) while the winner-distribution entropy moves from 0.13 to 1.00 in the 35B model and from 0.42 to 0.95 in the 122B model. The per-token spread stays put; the identity of each domain’s leading expert changes. That relabeling is the specialization signal.

A candidate cause. A plausible reading gives the negative results a principled cause: reading a short prompt is a fairly generic operation (tokenizing, parsing, building context) for which the router falls back on broadly useful experts, and domain-specific computation is recruited once the model commits to producing domain-specific content token by token. On that reading supplied-text routing misses domain structure

because it observes the model only during the regime that has yet to recruit the domain machinery. This is a hypothesis about the cause; the data locate the effect.

Implication for how specialization is measured and read. An analysis that characterizes expert-topic structure from prefill routing, or from routing pooled over a sequence without separating the regimes, sees the concentrated prefill picture and understates specialization. The same probe read over generation recovers per-domain structure. Two consequences follow. First, negative results obtained over supplied text, including the canonical Mixtral analysis (Jiang et al., 2024), report the prefill regime and are silent about generation; the demonstrated claim covers the Qwen3.5 lineage, and the reading of any particular prior result as understated is an inference from the measurement convention, since we re-measure none of them. Second, routing analyses that ask what experts specialize in should measure the two regimes separately and read specialization from generation; expert-identification methods, benchmark suites for specialization, and expert-naming pipelines that operate on supplied text inherit the prefill picture by construction. The result leaves open what the generation-regime experts encode, and analyses of prefill routing for their own sake are unaffected by it.

6 Limitations

Both checkpoints are 8-bit quantized community finetunes of Qwen3.5 MoE models; the pattern reproduces across two sizes with non-corresponding indices, which supports the method-level reading, but the demonstrated claim covers this lineage. Generated blocks are long and frequently reach the cap (in the 122B run 49 of 60 cells hit 2,048 tokens, mean generated length about 1,868), and every cell eventually spills into chat-template continuation; trimming at the spill boundary leaves the 35B winner list unchanged, which argues against a spill artifact, while generation budgets vary per prompt. The domain winner is a coarse statistic that treats a domain with two near-tied leaders and a domain with one dominant expert alike, and with three prompts per domain it is the least reliable layer of the analysis; the claims rest on the aggregate concentration contrasts alone; no single domain’s assignment is load-bearing. The concentration summaries are descriptive: no randomization null is attached to the 18-versus-20 and 13-versus-18 contrasts, although their size leaves little room for chance dispersion.

7 Conclusion

Two MoE language models route prompt tokens and generated tokens to different experts, and the difference is structured: over the prompt-reading regime a single generic expert wins almost every domain, and over the generation regime the domains acquire distinct leading experts. The shift is large, survives matching the regimes in token count, and reproduces across two model sizes with unrelated expert indices. Expert specialization measured over supplied text, the regime that is cheap and deterministic to capture, is therefore understated by construction, and the negative results built on such measurements are silent about the regime in which specialization is expressed. Analyses that ask what an MoE’s experts specialize in should separate the two regimes and read specialization from generation.

Data and Code Availability

The data supporting this paper accompany it in the `data/` directory, organized as three runs: the 35B 60-prompt primary probe, the 35B token-balanced three-chunk control, and the 122B replication (prefill and generation captures). Each run folder contains the per-domain winner tables from which every number in the paper is taken; `data/README.md` maps each run to the claims it supports, and `SOURCES.md` gives a claim-by-claim source index. The two figures and the concentration summaries in Table 1 are reproduced from the winner lists by `make_figures.py`, which re-runs no model. Raw router tensors (`.npz`) are included as non-load-bearing provenance. Two provenance choices are stated plainly: the 122B analysis was framed in its source notes around locating an analog of one particular expert, and we set that framing aside and report only the regime-dependent concentration the same tables support; and we use the reconstructed, log-derived routing tables in place of earlier hand-summarized figures. No new model training was performed.

AI Collaboration Statement

Generative AI (Anthropic’s Claude) was used as a tool in preparing this manuscript: to assist with drafting and revising prose, organizing the manuscript, and locating and formatting the bibliography. All experiments, captures, and numerical results are the author’s; every reported value was traced to a source capture (see `SOURCES.md`), and every reference was verified by the author against its authoritative record. No AI system is an author. The author reviewed the entire text and takes full responsibility for its content.

Funding and Competing Interests

This work received no external funding. The author declares no competing interests. The models studied are third-party fine-tunes used as-is for measurement; the author has no affiliation with their creators.

References

- Mynampati Sri Ranganadha Avinash. Task-conditioned routing signatures in sparse Mixture-of-Experts transformers. arXiv:2603.11114, 2026.
- Damai Dai, Chengqi Deng, Chenggang Zhao, R. X. Xu, Huazuo Gao, Deli Chen, Jiashi Li, Wangding Zeng, Xingkai Yu, Y. Wu, Zhenda Xie, Y. K. Li, Panpan Huang, Fuli Luo, Chong Ruan, Zhifang Sui, and Wenfeng Liang. DeepSeekMoE: Towards ultimate expert specialization in mixture-of-experts language models. arXiv:2401.06066, 2024.
- William Fedus, Barret Zoph, and Noam Shazeer. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research*, 23(120):1–39, 2022. doi: 10.48550/arXiv.2101.03961.
- Shun ichiro Hayashi, Daichi Mukunoki, Tetsuya Hoshino, and Takahiro Katagiri. Layer-wise MoE routing locality under shared-prefix code generation: Token-identity decomposition and compile-equivalent fork redundancy. arXiv:2604.17182, 2026.
- Robert A. Jacobs, Michael I. Jordan, Steven J. Nowlan, and Geoffrey E. Hinton. Adaptive mixtures of local experts. *Neural Computation*, 3(1):79–87, 1991. doi: 10.1162/neco.1991.3.1.79.
- Albert Q. Jiang, Alexandre Sablayrolles, Antoine Roux, Arthur Mensch, Blanche Savary, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Emma Bou Hanna, Florian Bressand, Gianna Lengyel, Guillaume Bour, Guillaume Lample, L  lio Renard Lavaud, Lucile Saulnier, Marie-Anne Lachaux, Pierre Stock, Sandeep Subramanian, Sophia Yang, Szymon Antoniak, Teven Le Scao, Th  ophile Gerv  t, Thibaut Lavril, Thomas Wang, Timoth  e Lacroix, and William El Sayed. Mixtral of experts. arXiv:2401.04088, 2024.
- Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles (SOSP)*, 2023. doi: 10.1145/3600006.3613165.
- Noam Shazeer, Azalia Mirhoseini, Krzysztof Mazi  rz, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Outrageously large neural networks: The sparsely-gated Mixture-of-Experts layer. arXiv:1701.06538, 2017.
- An Yang et al. Qwen3 technical report. arXiv:2505.09388, 2025.
- Yinmin Zhong, Shengyu Liu, Junda Chen, Jianbo Hu, Yibo Zhu, Xuanzhe Liu, Xin Jin, and Hao Zhang. DistServe: Disaggregating prefill and decoding for goodput-optimized large language model serving. In *18th USENIX Symposium on Operating Systems Design and Implementation (OSDI 24)*, 2024. doi: 10.48550/arXiv.2401.09670.
- Barret Zoph, Irwan Bello, Sameer Kumar, Nan Du, Yanping Huang, Jeff Dean, Noam Shazeer, and William Fedus. ST-MoE: Designing stable and transferable sparse expert models. arXiv:2202.08906, 2022.